

Drucker–Prager Constitutive Update for Static Two-Phase UL-MPM

2026-05-11

1. Scope

This note derives the hand-written gradient and Hessian ingredients needed to add a Drucker–Prager solid constitutive law to `src/igampm/MPM/static_twophase_ulmpm.py`. The target use is the same static two-phase updated-Lagrangian MPM formulation documented in `docs/GeoWarp-main/Implicit`. The Drucker–Prager reference logic is cross-checked against `docs/GeoWarp-main/problems/material_tests/t` but the present note rewrites the model in invariant tensor form and gives explicit hand-derived Jacobian/Hessian expressions. No auto-differentiation is used.

The implementation target is perfect plasticity with isotropic small-strain elasticity. Hardening is not included. The effective stress is updated; the total stress in the two-phase momentum balance remains

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}' - p_f \mathbf{I}.$$

2. Elastic Predictor

Let the committed elastic strain at the start of the load step be

$$\boldsymbol{\varepsilon}_e^n.$$

At one Newton iterate of the global static solve, the current total incremental strain is

$$\Delta \boldsymbol{\varepsilon} = \frac{1}{2} \left(\nabla \mathbf{u} + \nabla \mathbf{u}^T \right).$$

The elastic trial strain is

$$\boldsymbol{\varepsilon}^{\text{tr}} = \boldsymbol{\varepsilon}_e^n + \Delta \boldsymbol{\varepsilon}.$$

With Lamé constants (λ, μ) , define the isotropic elastic tensor

$$\mathbb{C}_e : \mathbf{a} = \lambda \operatorname{tr}(\mathbf{a}) \mathbf{I} + 2\mu \mathbf{a}.$$

The trial effective stress is

$$\boldsymbol{\sigma}^{\text{tr}} = \mathbb{C}_e : \boldsymbol{\varepsilon}^{\text{tr}}.$$

In three dimensions, the mean pressure and deviatoric stress are

$$p = \frac{1}{3} \operatorname{tr}(\boldsymbol{\sigma}), \quad \mathbf{s} = \boldsymbol{\sigma} - p \mathbf{I}.$$

The equivalent deviatoric magnitude used below is

$$t = \|\mathbf{s}\| = \sqrt{\mathbf{s} : \mathbf{s}}, \quad q = \sqrt{\frac{3}{2}} t.$$

For the trial state we write $(p^{\text{tr}}, \mathbf{s}^{\text{tr}}, t^{\text{tr}})$.

3. Drucker–Prager Yield Surface and Plastic Potential

For the yield surface, define

$$A_f = \frac{2\sqrt{6}c \cos \phi}{3 - \sin \phi}, \quad B_f = \frac{2\sqrt{6} \sin \phi}{3 - \sin \phi}, \quad k_f = \eta A_f,$$

where c is cohesion, ϕ is friction angle, and η is the smoothing `shape_factor`.

For the plastic potential, define

$$A_g = \frac{2\sqrt{6}c \cos \psi}{3 - \sin \psi}, \quad B_g = \frac{2\sqrt{6} \sin \psi}{3 - \sin \psi}, \quad k_g = \eta A_g,$$

where ψ is dilation angle.

The yield function and plastic potential are written as

$$f(\boldsymbol{\sigma}) = \sqrt{t^2 + k_f^2} - A_f + B_f p,$$

$$g(\boldsymbol{\sigma}) = \sqrt{t^2 + k_g^2} - A_g + B_g p.$$

This is algebraically equivalent to the GeoWarp expression

$$\sqrt{\frac{2}{3}q^2 + k_f^2} - (A_f - B_f p),$$

because $\frac{2}{3}q^2 = t^2$.

4. First Derivatives

Let

$$r_f = \sqrt{t^2 + k_f^2}, \quad r_g = \sqrt{t^2 + k_g^2}.$$

Using $\partial p / \partial \boldsymbol{\sigma} = \mathbf{I}/3$ and

$$\frac{\partial t}{\partial \boldsymbol{\sigma}} = \frac{\partial t}{\partial \mathbf{s}} : \frac{\partial \mathbf{s}}{\partial \boldsymbol{\sigma}} = \frac{\mathbf{s}}{t} : \mathbb{P}_{\text{dev}} = \frac{\mathbf{s}}{t},$$

the gradients are

$$\frac{\partial f}{\partial \boldsymbol{\sigma}} = \frac{B_f}{3} \mathbf{I} + \frac{\mathbf{s}}{r_f}, \quad \frac{\partial g}{\partial \boldsymbol{\sigma}} = \frac{B_g}{3} \mathbf{I} + \frac{\mathbf{s}}{r_g}.$$

Define

$$\mathbf{n}_f = \frac{\mathbf{s}}{r_f}, \quad \mathbf{n}_g = \frac{\mathbf{s}}{r_g}.$$

Then

$$\nabla_{\boldsymbol{\sigma}} f = \frac{B_f}{3} \mathbf{I} + \mathbf{n}_f, \quad \nabla_{\boldsymbol{\sigma}} g = \frac{B_g}{3} \mathbf{I} + \mathbf{n}_g.$$

5. Hessian of the Plastic Potential

The deviatoric projector is

$$\mathbb{P}_{\text{dev}} : \mathbf{a} = \mathbf{a} - \frac{1}{3} \text{tr}(\mathbf{a}) \mathbf{I}.$$

Since k_g is constant,

$$\delta \mathbf{n}_g = \delta \left(\frac{\mathbf{s}}{r_g} \right) = \frac{\delta \mathbf{s}}{r_g} - \frac{\mathbf{s}}{r_g^2} \delta r_g.$$

Because

$$\delta \mathbf{s} = \mathbb{P}_{\text{dev}} : \delta \boldsymbol{\sigma}, \quad \delta r_g = \frac{\mathbf{s} : \delta \mathbf{s}}{r_g} = \frac{\mathbf{s} : \delta \boldsymbol{\sigma}}{r_g},$$

we obtain the Hessian action

$$\delta(\nabla_{\sigma} g) = \mathbb{H}_g : \delta \boldsymbol{\sigma} = \frac{1}{r_g} \mathbb{P}_{\text{dev}} : \delta \boldsymbol{\sigma} - \frac{\mathbf{s} : \delta \boldsymbol{\sigma}}{r_g^3} \mathbf{s}.$$

Equivalently, in fourth-order tensor form,

$$\mathbb{H}_g = \frac{1}{r_g} \mathbb{P}_{\text{dev}} - \frac{1}{r_g^3} \mathbf{s} \otimes \mathbf{s}.$$

The same pattern holds for the yield function:

$$\mathbb{H}_f = \frac{1}{r_f} \mathbb{P}_{\text{dev}} - \frac{1}{r_f^3} \mathbf{s} \otimes \mathbf{s}.$$

In the present implementation, only $\mathbb{H}_g$ is needed explicitly inside the local Newton Jacobian.

6. Return Mapping

The backward-Euler plastic update is

$$\boldsymbol{\varepsilon}_e - \boldsymbol{\varepsilon}^{\text{tr}} + \Delta \lambda \frac{\partial g}{\partial \boldsymbol{\sigma}} = \mathbf{0}, \quad f(\boldsymbol{\sigma}) = 0, \quad \boldsymbol{\sigma} = \mathbb{C}_e : \boldsymbol{\varepsilon}_e.$$

Using the volumetric/deviatoric split and isotropic elasticity, the update becomes a two-scalar problem. Let

$$K = \lambda + \frac{2}{3}\mu.$$

Then

$$p = p^{\text{tr}} - KB_g \Delta \lambda.$$

For the deviatoric part,

$$\mathbf{s} = \mathbf{s}^{\text{tr}} - 2\mu \Delta \lambda \frac{\mathbf{s}}{r_g}.$$

Hence $\mathbf{s}$ stays coaxial with $\mathbf{s}^{\text{tr}}$, so only its magnitude t needs to be solved:

$$t - t^{\text{tr}} + 2\mu \Delta \lambda \frac{t}{r_g} = 0.$$

Together with the yield condition,

$$r_f - A_f + B_f (p^{\text{tr}} - KB_g \Delta \lambda) = 0,$$

the local unknowns are just $(t, \Delta \lambda)$.

Define

$$R_1(t, \Delta \lambda) = t - t^{\text{tr}} + 2\mu \Delta \lambda \frac{t}{r_g},$$

$$R_2(t, \Delta \lambda) = r_f - A_f + B_f (p^{\text{tr}} - KB_g \Delta \lambda).$$

The local Newton Jacobian is

$$\mathbf{J}_{\text{loc}} = \begin{bmatrix} \frac{\partial R_1}{\partial t} & \frac{\partial R_1}{\partial \Delta \lambda} \\ \frac{\partial R_2}{\partial t} & \frac{\partial R_2}{\partial \Delta \lambda} \end{bmatrix},$$

with entries

$$\begin{aligned} \frac{\partial R_1}{\partial t} &= 1 + 2\mu \Delta \lambda \frac{k_g^2}{r_g^3}, & \frac{\partial R_1}{\partial \Delta \lambda} &= 2\mu \frac{t}{r_g}, \\ \frac{\partial R_2}{\partial t} &= \frac{t}{r_f}, & \frac{\partial R_2}{\partial \Delta \lambda} &= -B_f KB_g. \end{aligned}$$

7. Stress Reconstruction

Once $(t, \Delta\lambda)$ is known,

$$p = p^{\text{tr}} - KB_g \Delta\lambda.$$

If $t^{\text{tr}} > 0$, define

$$\alpha = \frac{t}{t^{\text{tr}}}, \quad \mathbf{s} = \alpha \mathbf{s}^{\text{tr}}.$$

Then

$$\boldsymbol{\sigma} = p\mathbf{I} + \mathbf{s}.$$

The updated elastic strain follows from Hooke's law, or equivalently from the plastic flow relation

$$\boldsymbol{\varepsilon}_e = \boldsymbol{\varepsilon}^{\text{tr}} - \Delta\lambda \left(\frac{B_g}{3} \mathbf{I} + \frac{\mathbf{s}}{r_g} \right).$$

8. Consistent Tangent Action

The global MPM assembly only needs the action of the algorithmic tangent on one trial strain perturbation $\delta\boldsymbol{\varepsilon}$.

First,

$$\delta p^{\text{tr}} = K \operatorname{tr}(\delta\boldsymbol{\varepsilon}), \quad \delta \mathbf{s}^{\text{tr}} = 2\mu \operatorname{dev}(\delta\boldsymbol{\varepsilon}).$$

If $t^{\text{tr}} > 0$, define the unit trial deviatoric direction

$$\mathbf{n}^{\text{tr}} = \frac{\mathbf{s}^{\text{tr}}}{t^{\text{tr}}}, \quad \delta t^{\text{tr}} = \mathbf{n}^{\text{tr}} : \delta \mathbf{s}^{\text{tr}}.$$

Differentiate the local system:

$$\begin{aligned} \frac{\partial R_1}{\partial t} \delta t + \frac{\partial R_1}{\partial \Delta\lambda} \delta(\Delta\lambda) &= \delta t^{\text{tr}}, \\ \frac{\partial R_2}{\partial t} \delta t + \frac{\partial R_2}{\partial \Delta\lambda} \delta(\Delta\lambda) &= -B_f \delta p^{\text{tr}}. \end{aligned}$$

Therefore

$$\begin{bmatrix} \delta t \\ \delta(\Delta\lambda) \end{bmatrix} = \mathbf{J}_{\text{loc}}^{-1} \begin{bmatrix} \delta t^{\text{tr}} \\ -B_f \delta p^{\text{tr}} \end{bmatrix}.$$

The pressure variation is

$$\delta p = \delta p^{\text{tr}} - KB_g \delta(\Delta\lambda).$$

For the deviatoric stress, using $\mathbf{s} = \alpha \mathbf{s}^{\text{tr}}$ with $\alpha = t/t^{\text{tr}}$,

$$\delta \mathbf{s} = \alpha \delta \mathbf{s}^{\text{tr}} + \mathbf{n}^{\text{tr}} (\delta t - \alpha \delta t^{\text{tr}}).$$

Hence the full consistent tangent action is

$$\delta \boldsymbol{\sigma} = \delta p \mathbf{I} + \alpha \delta \mathbf{s}^{\text{tr}} + \mathbf{n}^{\text{tr}} (\delta t - \alpha \delta t^{\text{tr}}).$$

This is the hand-written algorithmic tangent used by the global Newton solve.

9. Remarks on the Implemented Branch

- The constitutive model is perfect Drucker–Prager plasticity with no hardening.
- The algorithm above is derived for the smooth branch of the cone and avoids any auto-diff construction.
- To stay aligned with the current GeoWarp reference script, the tension-side tip branch $p^{\text{tr}} \geq 0$ is not given a separate apex return in the present implementation; that branch falls back to the elastic predictor.
- The implementation is intended for the small-strain constitutive path of `StaticTwoPhaseULMPM`. It is not used by the finite-strain `neoHookean` branch.
- The local solver uses a fixed-iteration Newton loop with a small tolerance, then the global Jacobian/Hessian is checked by finite differences.